

1. AI & Vision

Primary Focus: Model Studio API connection, system prompt engineering, and core inference script.

- **Task 1: Model Studio Connection**

- Obtain your DASHSCOPE_API_KEY from Alibaba Cloud Model Studio console.
- Set up a clean virtual environment: pip install openai pydantic pillow.

- **Task 2: Build ai_engine.py**

- Create the standalone script connecting to qwen2.5-vl-72b-instruct (or 7b).
- Enforce strict JSON output with two payloads:
 1. Extracted OCR metadata (gtin, batch_number, expiry_date, drap_reg_num).
 2. Print anomaly evaluation (print_score, defects, bbox_2d).

- **Task 3: Implement Dual-Gate Scoring Logic**

- Write the Python scoring function:

$$S_{\text{final}} = S_{\text{DB}} \times (0.60 \cdot S_{\text{rule}} + 0.40 \cdot S_{\text{visual}})$$

- **Tonight's Deliverable:** A working Python script that takes a local image of a medicine box, runs inference against Qwen2.5-VL, and prints the formatted JSON output with bounding boxes.

2. Mobile App Developer (Flutter)

Primary Focus: Camera viewfinder with zoom lock and screen scaffolding.

- **Task 1: Project Initialization & Dependencies**

- Create a new Flutter project (dawae_check_app) and push the initial commit to GitHub.
- Add dependencies to pubspec.yaml: camera, http, image_picker, permission_handler.

- **Task 2: Camera Viewfinder with 3x Macro Lock**

- Implement CameraScreen with a custom overlay showing an alignment guide reticle.
- Add zoom toggle buttons (1x, 2x, 3x) with 3x selected as the default inspection mode.
- **Task 3: Result Screen Wireframe**
 - Scaffold ResultScreen containing:
 - Status Header (Green **GENUINE** / Red **COUNTERFEIT** badge).
 - Score progress indicator (0 – 100%).
 - Image preview placeholder ready for defect bounding-box overlays.
- **Tonight's Deliverable:** Flutter app running on an Android/iOS emulator or test device that opens the camera, locks focus/3x zoom, captures a photo, and navigates to the result screen.

3. Cloud & Backend Engineer (FastAPI / Qoder IDE)

Primary Focus: Project scaffolding, database migration, and mock seed records.

- **Task 1: Backend Scaffolding with Qoder IDE**
 - Initialize the FastAPI project structure (app/api, app/core, app/models, app/schemas).
 - Configure environment handling with python-dotenv and Pydantic Settings.
- **Task 2: Database Setup (PostgreSQL / Supabase)**
 - Create the database instance and run migrations for the 3 core tables:
 - manufacturers (id, company_name, drap_license_num).
 - batch_registry (id, gtin, brand_name, batch_number, official_expiry_date, drap_reg_num).
 - scanned_logs (id, device_id, facility_id, scanned_gtin, batch_number, verdict, authenticity_score, created_at).
- **Task 3: Seed Database & Build Stub API**
 - Insert seed records for 4 medicines: Augmentin (valid), Panadol (expired), Arinac (unregistered), Brufen (valid DB/fake print).

- Implement the POST /api/v1/verify-packaging endpoint returning a mock JSON response.
- **Tonight's Deliverable:** A running local FastAPI server accessible at <http://localhost:8000/docs> where the verify endpoint accepts a multipart image upload and queries the database.

4. UI/UX Designer & Pitch/Dataset Lead

Primary Focus: Figma wireframes, physical test dataset collection, and pitch structure.

- **Task 1: Figma UI Prototype**
 - Design clean mobile wireframes for 3 core screens:
 1. **Camera Screen:** Viewfinder guide frame, lighting tip banner, 1x/2x/3x toggle.
 2. **Analysis Screen:** Pulsing radar/scanning animation.
 3. **Result Screen:** Color-coded status badge, Layer 1 DB checks, Layer 2 print quality breakdown with defect tags.
- **Task 2: Physical Packaging Dataset Collection**
 - Collect 3 real medicine boxes (e.g., Panadol, Augmentin, Arinac).
 - Take clear, high-resolution 3x macro photos of the batch numbers, expiry dates, and 2D DataMatrix codes.
 - Print 2 mock fake labels on a standard home/office inkjet printer on cheap paper to create test samples with visible halftone dithering.
- **Task 3: Pitch Deck Outline**
 - Set up a Google Slides or Canva presentation with the standard 10-slide hackathon structure (Problem, Solution, Tech Stack, Dual-Gate Architecture, Market & Alkhidmat Impact).
- **Tonight's Deliverable:** Figma link with the 3 main screens + a shared Google Drive folder containing at least 6 macro test photos (real + inkjet-printed samples).